REPOSITORY ASSESSMENT

Documentation compliance audit

Review date: 7 November 2026. Scope: current repository source and available automated checks. Application behavior, database schema deployment, and client acceptance were not verified in a running environment. No application code or business specification was changed.

Verdict

The system does not yet meet the complete requirements of both documents. The main buy, receive, reserve, pack, transfer, sell, and deliver workflows are implemented in source. Several required modules remain placeholders, enterprise governance features are absent, and dashboard calculations have correctness gaps.

The documents describe different levels of scope:

- System flow, all 26 sections: includes an explicit implementation index distinguishing working workflows from planned modules. The repository broadly matches those stated limits; this does not make the whole target system complete.
- KFF System Documentation PDF, version 1.0, 7 November 2026, all 23 pages: describes a broader enterprise target, including formula revisions, approvals, count sessions, configuration, and governance. Page 1 explicitly says it is a functional blueprint rather than evidence of deployment, and page 23 identifies enterprise governance as intended target procedures.

"Implemented" below means source support was found, with applicable request/schema tests passing. It does not mean production acceptance or database concurrency was demonstrated.

Confirmed gaps and defects

1. Packing formulas cannot be maintained through the application

Priority: high. System-flow sections 4 and 22; PDF sections 4, 7 and 18, pages 6, 8-9 and 17.

The Packing Formulas page is a placeholder. There is no BOM maintenance router in application composition. Packing calculations read existing rows, so new product/formula setup cannot be completed through the documented UI.

The BOM schema stores parent, component, quantity, unit and audit fields, but has no formula revision, release approval, effective date or revision-note model. Historical order snapshots are implemented; approved formula revision management from the PDF is not.

Required completion: BOM editor/API and validation; separately define and implement the PDF's formula release/revision lifecycle while retaining order snapshots.

2. Dedicated reports are absent

Priority: high. System-flow section 18; PDF section 12, pages 12-13, and acceptance scenario 13.

The Inventory Reports page has five tabs displaying planned-report placeholders. There is no reporting API in application composition, and no dedicated packing, purchasing or transfer report pages.

Stock, movements and dashboard screens supply operational information, but do not fulfill the documented report set, including material consumption, packing variance, supplier purchases, lead time, fulfillment rate, adjustment trends and lot genealogy reporting.

Required completion: operational report queries, filters, exports and source-document drill-down, with quantities grouped by compatible primary units.

3. Dashboard totals combine incompatible units

Priority: high; defect in an existing feature. System-flow sections 2-3; PDF sections 8 and 12, pages 9 and 12.

In dashboard calculations, raw and finished quantities are added directly, and type/location buckets sum every stock quantity without grouping or converting UOMs. Overview cards label the resulting quantities generically as "units."

Example: a location containing 50 kg of spice and 500 pouches contributes 550 to one quantity total. Even 1 kg plus 500 g becomes 501 if those are the items' respective primary units. These totals have no valid common physical meaning.

Required correction: group by primary UOM, or explicitly convert compatible classes to a documented common unit. Keep inventory value totals separate from physical quantities.

4. Dashboard active/pending lists can show closed work and hide open work

Priority: medium; defect in an existing feature. System-flow section 2; PDF sections 12-13.

The packing, purchase-order and transfer list queries in dashboard calculations sort recent records and apply limits without filtering their statuses. Separate summary-count queries do filter statuses. The frontend renders those results as Active Operations and Active Transfers.

Consequently, recent Completed/Cancelled records can occupy the list, while an older outstanding order falls outside its limit; the displayed cards can also disagree with the summary counts.

Required correction: filter the list queries to the intended open states before sorting/limiting, and align each list's definition with its summary count.

5. Low-stock alerts omit items without a stock row

Priority: medium; defect in an existing feature. System-flow section 2; PDF section 12.

The low-stock candidates in dashboard calculations are built exclusively from existing inventory balances. An active material that has never been received has no balance row, so its zero availability produces no low-stock alert. Stock with failed QC or expiry also contributes to the numeric availability used for this alert, although it cannot be allocated for normal packing.

Required correction: start with the applicable active item catalog, include zero-stock items, and distinguish numeric availability from eligible supply. Define item/UOM-specific reorder policies.

6. Counts support immediate corrections, not enterprise count sessions

Priority: high for the PDF target; documented current limitation in the system flow. System-flow sections 2 and 14; PDF section 8, page 10, and sections 17 and 22.

The Stock Counts page uses the stock-adjustment form and service. Count rules enforce an expected balance and reservation protection, but reject an unchanged count. There are no count-session, observation, variance-review or supervisor-approval records in the schema.

Required completion: session scope, captured observations including unchanged counts, variance review, approval, and count-linked adjustment posting. Preserve existing stale-balance and reserved-stock checks.

7. QC disposition, scrap and exception approvals are incomplete

Priority: high for the PDF target. System-flow sections 2, 13 and 23; PDF sections 6, 8, 11 and 17, and acceptance scenario 14.

Lot QC status/date editing and reservation guards exist in the lot service. However, the lot update contract has no QC decision notes/evidence, and stores current metadata rather than a separate QC decision history.

There is no dedicated scrap/disposal API or reason-code maintenance API in application composition. The stock-adjustment operation uses reasons in the Inventory Adjustment category. Existing internal transfers are restricted to finished goods from warehouses to Warehouse/Branch/Store destinations; they do not provide the general raw-material Quarantine/Scrap movement path described in the PDF.

There is no maker/checker approval model for formula release, sensitive master changes, large adjustments or write-offs in the schema. Broad role permissions and authenticated actors do not implement separate preparation and approval.

Required completion: define the approved exception scope, add decision/evidence records and approval enforcement, then expose controlled disposition and stock movements that preserve reservations and ledger consistency.

8. Brands and company configuration remain placeholders

Priority: medium. System-flow sections 2 and 21; PDF sections 4, 11-12 and 18.

The Brands page and Company Settings page are placeholders. Their storage and APIs are absent; items have no brand relationship in the schema.

The dashboard hard-codes a low-stock threshold of 10 and an expiry horizon of 30 days in its service, while currency formatting hard-codes PHP. This falls short of the PDF's controlled thresholds, currency/valuation policy, approver assignments and numbering configuration. Item-specific packaging records exist in the schema but have no maintenance API.

Coverage of all system-flow sections

Sections	Assessment	Evidence / qualification
1. Purpose	Substantially supported in source	Core operational domains exist; complete reporting and master-data setup do not.
2. Modules and dashboard	Partial	Working operational modules and explicit planned-module index; dashboard defects above.
3. Master data	Implemented core scope	Items, suppliers, customers, UOM, and locations have APIs, UI and reference guards.
4. BOM	Partial	Packing service calculates requirements and limiting capacity; formula maintenance is absent.
5. Purchasing	Implemented in source	PO service supports draft editing, send/cancel and line quantities; receiving updates fulfillment.
6. Receiving	Implemented in source	Receiving service locks PO context, supports batch/partial receipts, checks stale/over-receipts, creates linked lots, ledger and balances in one transaction.
7. Lots	Implemented core scope	Lot service exposes purchase/packing provenance, optional dates, QC and stock/movement detail; broader PDF disposition workflow is incomplete.
8. Warehouse 1	Implemented shared model	Item/lot/location balances and material eligibility are used; example warehouse codes are configured data.
9. Packing orders	Implemented in source	Packing service enforces Draft/Planned/In Progress/Completed/Cancelled transitions.
10. Material snapshots	Implemented in source	Creation copies primary-UOM requirements into order materials; historical operations use those rows.
11. Reservations	Implemented in source; live safety unverified	Packing allocation uses eligible lots and locks; availability includes outstanding active packing and delivery reservations.
12. Packing completion	Implemented in source; live safety unverified	One transaction consumes proportional inputs, releases allocations, creates output lot/Produce entry, and updates order.
13. Stock and ledger	Implemented core scope	Stock service and workflow services maintain balances with new ledger entries. No runtime ledger update/delete path was found. Dedicated Scrap posting is absent.
14. Adjustments	Implemented core scope	Expected-balance checks, reason, actor, reservation protection and transactional ledger delta exist. PDF count sessions are additional work.

Sections	Assessment	Evidence / qualification
15-16. Transfers and In Transit	Implemented in source; live safety unverified	Transfer service dispatches paired movements, recovers original transit from ledger, receives increments and completes after shipped quantities arrive.
17. Branch/store, sales and delivery	Implemented in source; live safety unverified	Shared stock model; sales confirmation does not reserve stock; delivery allocates demand, protects transit and posts Sale/fulfillment atomically.
18. Reports	Missing target workflows	Report placeholders and no reporting API.
19. Users and roles	Implemented core scope	Backend permission checks; sessions load current role settings; access policy protects the last active user manager.
20. Audit fields	Implemented core scope	Authenticated actors and timestamps populate operational writes. Separate approval/evidence histories required by the PDF are absent.
21. Navigation	Largely aligned	Sidebar follows the documented groups; planned destinations remain placeholders.
22. Complete flow	Partial	Core operations exist, but application-based BOM setup and reporting prevent complete target coverage.
23. Business rules	Substantially supported in source	Exact quantities, snapshots, provenance, reservation checks, distinct locations and foreign keys are implemented; runtime acceptance remains outstanding.
24. Architecture	Aligned	Next.js feature hooks -> Express validation/controller -> services -> Drizzle/PostgreSQL.
25. Transaction safety	Source support; not proven live here	Transaction/lock code and opt-in integration suites exist; PostgreSQL suites skipped in this review.
26. Design principle	Broadly aligned	One inventory model and feature services support the core business; planned modules remain incomplete.

PDF acceptance scenarios

These are source assessments, not signed UAT results. Scenario numbers refer to PDF section 22, pages 21-22.

IDs	Assessment
01. Supplier offers	Source supports multiple supplier offers and preferred-offer protection.
02-03. Batch receiving / over-receipt	Service and rules support the required behavior; no live receiving transaction/concurrency test was run.
04-06. Packing shortage / start / below-target completion	Source and tests support intended behavior; real PostgreSQL integration suite skipped.
07. Count below reservation	Source and rule tests enforce rejection; live database behavior not demonstrated.
08. Incremental transfer receiving	Source support; PostgreSQL integration suite skipped.
09-10. Sales allocation / customer completion	Source support; PostgreSQL integration suite skipped.
11. Permission revocation	Session lookup reloads persisted permissions; HTTP tests passed, but role/user PostgreSQL integration suite skipped.
12. Backward lot tracing	Provenance, order snapshots, reservations and movements store the links; end-to-end client drill-through was not demonstrated. A dedicated genealogy report is absent.
13. Low-stock / expiry / transit review	Partial: dashboard defects, fixed alert policies and missing reports prevent full acceptance.
14. Count and controlled disposal	Partial: immediate count adjustments exist; count-session evidence, dedicated disposal and separate approvals are absent.

Verification performed and limits

Check	Result
Read PDF text	All 23 pages extracted and reviewed, including target/governance qualifications.
Read system flow	All 26 sections reviewed, with application instructions and packing/transfer/sales-delivery contracts.
Backend `npm test`	195 tests: 190 passed, 5 skipped, 0 failed.
Backend `npm run typecheck`	Passed.
Frontend `npm run test:api`	209 passed, 0 skipped, 0 failed.
Frontend installed TypeScript `tsc --noEmit`	Passed.
PostgreSQL acceptance	Not run. Skipped suites cover packing, internal transfers, sales/delivery, roles/users, and sample import/rebuild; they require `TEST_DATABASE_URL`. No operational database was used as a fallback.
Browser / deployed environment	Not tested. No live UI workflow, migration state, ledger reconciliation, production configuration or deployment acceptance claim is made.
Business continuity	Backup/restore, monitoring and support readiness from PDF section 22 require environment and operational evidence; unverified here.

Most HTTP tests use fake services and frontend tests mock requests. Passing them verifies contracts and guards, not real database rollback or locking. Existing integration test definitions are useful evidence of intended verification, but skipped tests do not establish acceptance. Builds were not run for this review because application code was unchanged.

Recommended completion order

1. Correct dashboard unit totals, status filtering and zero-stock/eligibility alerts.
2. Deliver BOM maintenance so operational master data can be configured through the application; include revision/approval behavior if accepting the PDF target.
3. Implement dedicated reports and company settings, then brands and remaining master-data maintenance.
4. Implement count sessions, QC evidence, disposition movements and maker/checker controls for the agreed enterprise scope.
5. Run PostgreSQL integration tests, add missing real receiving/count verification, demonstrate PDF acceptance scenarios with configured data, and confirm migration, reconciliation and operational readiness.

Keep the canonical system flow as the shared specification. If the PDF's additional enterprise requirements become implementation scope, extend its relevant existing sections and module contracts explicitly; do not interpret the PDF's target procedures as already deployed capabilities.